

BATUMI, Georgia

WMO: 374840

Lat: 41.610N

Lon: 41.600E

Elev: 7

StdP: 101.25

Time Zone: 4.00 (E04)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.1	0.9	-6.8	2.1	7.0	-5.2	2.4	7.1	14.4	9.5	13.4	9.8	5.0	140	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.3	30.0	24.8	29.0	24.5	28.1	23.8	26.0	29.1	25.3	28.3	24.5	27.4	3.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.1	28.7	24.1	19.0	27.6	23.4	18.2	26.8	80.8	29.0	76.9	28.5	74.2	27.5	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.3	11.2	10.1	-2.4	34.1	1.9	1.8	-3.8	35.4	-4.9	36.5	-5.9	37.5	-7.3	38.8		
			-3.3	26.6	2.1	0.7	-4.8	27.1	-6.0	27.5	-7.2	27.9	-8.7	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	8.2	8.5	10.4	13.0	17.2	21.4	23.5	24.6	21.4	17.5	13.2	9.9
	DBStd	6.62	3.95	4.00	4.33	3.79	3.16	2.21	1.89	1.89	2.45	2.96	3.66	4.17
	HDD10.0	277	82	72	47	9	0	0	0	0	0	1	13	54
	HDD18.3	1540	313	276	248	166	59	2	0	0	6	51	157	262
	CDD10.0	2391	27	29	61	99	223	342	419	454	343	232	109	51
	CDD18.3	612	0	0	3	7	23	94	161	196	99	24	3	1
	CDH23.3	3429	0	1	30	54	92	391	953	1409	429	61	7	2
	CDH26.7	613	0	0	6	12	29	49	152	314	40	9	0	0
Wind	WSAvg	4.5	6.0	5.1	4.8	4.0	3.4	3.5	3.5	3.5	4.1	4.5	5.5	6.1
Precipitation	PrecAvg	2575	251	198	154	109	91	170	171	229	294	291	285	305
	PrecMax	3493	405	419	266	311	187	332	374	544	506	597	645	582
	PrecMin	2016	94	30	54	20	28	73	72	35	97	9	11	110
	PrecStd	319	94	96	52	59	43	61	74	128	102	141	159	104
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.0	21.4	27.0	28.4	30.2	30.2	30.2	31.2	29.1	28.1	24.2	22.8
		MCWB	10.6	11.6	14.6	16.2	20.5	22.9	25.7	26.5	24.3	19.5	14.6	12.2
	2%	DB	17.2	18.2	22.9	24.5	25.1	27.9	29.1	30.1	27.9	24.9	21.9	19.9
		MCWB	9.0	10.7	13.1	15.4	19.4	23.8	24.9	25.8	23.2	19.1	13.9	10.7
	5%	DB	15.2	16.1	20.0	21.6	23.1	26.8	28.2	29.1	26.8	23.2	19.8	17.6
		MCWB	8.3	9.3	11.8	14.2	18.6	23.0	24.3	25.2	22.6	18.7	13.5	10.1
	10%	DB	13.2	14.1	17.1	18.9	21.9	25.6	27.2	28.4	25.9	21.9	18.1	15.8
		MCWB	7.8	8.5	10.9	13.3	18.1	22.2	23.6	24.5	22.0	18.1	12.8	9.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	12.9	15.4	17.2	22.0	25.2	26.6	26.8	24.9	21.4	17.1	14.1
		MCDB	15.9	19.0	24.7	24.8	26.3	28.0	29.4	30.3	28.3	24.6	20.7	17.2
	2%	WB	10.5	11.2	13.6	15.8	20.3	24.2	25.7	26.2	23.8	20.3	15.6	12.8
		MCDB	14.2	16.4	21.6	22.7	23.8	27.3	28.6	29.5	26.9	23.7	19.0	16.4
	5%	WB	9.7	10.2	12.3	14.8	19.4	23.3	24.9	25.7	23.0	19.3	14.7	11.6
		MCDB	13.3	14.2	18.6	19.9	22.6	26.3	27.7	28.8	26.0	22.5	18.0	15.4
	10%	WB	8.9	9.4	11.2	13.8	18.5	22.4	24.1	25.0	22.2	18.4	13.7	10.6
		MCDB	12.8	12.7	16.1	18.0	21.4	25.0	26.9	28.0	25.2	21.3	17.1	14.4
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.7	7.4	7.2	6.2	6.3	5.9	6.3	6.7	6.7	6.9	6.4
		MCDBR	8.4	9.8	12.7	12.4	9.1	7.7	6.7	6.8	7.5	8.3	9.1	8.5
		MCWBR	4.6	5.1	5.9	6.0	5.8	5.3	4.6	4.6	5.4	5.9	5.8	4.8
	5% WB	MCDBR	7.2	8.6	11.2	10.6	7.9	7.0	6.4	6.5	7.0	7.6	7.7	7.1
		MCWBR	4.5	4.7	5.4	5.6	5.5	5.0	4.5	4.5	5.4	5.7	5.8	4.7
Clear-Sky Solar Irradiance	taub	0.353	0.387	0.443	0.496	0.462	0.443	0.470	0.481	0.422	0.419	0.368	0.345	
	taud	2.284	2.175	2.061	1.991	2.181	2.306	2.264	2.223	2.369	2.294	2.376	2.344	
	Ebn at Noon	773	798	794	780	821	836	809	787	812	760	756	751	
	Edh at Noon	91	118	149	171	145	128	133	135	109	105	84	80	
All-Sky Solar Radiation	RadAvg	1.60	2.37	3.28	4.53	5.42	5.88	5.56	5.06	4.25	2.89	2.00	1.43	
	RadStd	0.21	0.28	0.30	0.53	0.46	0.49	0.56	0.43	0.38	0.32	0.29	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-110	+96	+38	

Nomenclature: See separate page